

Tianwei Xi

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[Portfolio](#) | [LinkedIn](#)

Education

Mt. San Antonio College | Walnut, CA

Transfer Coursework | January 2022 - June 2026

- Involvement: Dual Enrollment Program, Honors Program, Engineering Club, Rocketry Team
- Relevant Coursework: MATLAB, Circuits, Statics, Material Properties, Differential Equations.

University of California, Riverside | Riverside, CA

B.S. Mechanical Engineering | September 2026 - Present

Extracurriculars and Projects

Engineering Club Social Media Manager & Inter Club Council Representative

Mt. SAC | August 2023 - June 2024

- Social Media Manager: Managed the club's online and campus presence through Instagram, Discord, and posters, promoting technical, professional-development, and social events that regularly attracted 50+ attendees.
- Inter-Club Council Representative: Coordinated collaborations with organizations including the Mathematics Club, Coding Club, Society of Women Engineers, Astronomy Club, and many more to organize joint events and expand student engagement.
- Event Leadership: Planned and executed club events including a paper airplane engineering competition and game day, each drawing 50+ participants.

Vehicle Performance Modeling & Data Analysis

Mt. SAC | August 2024 - December 2024

- Collected and analyzed vehicle performance data across 9 experimental trials, modeling position, velocity, acceleration, and fuel consumption in MATLAB.
- Extrapolated experimental data to model performance on the Irwindale track, predicting a ~23 m/s top speed, 5 m/s² acceleration, and 0.15 gal fuel consumption over 50 seconds.

Stirling Engine Efficiency Analysis

Mt. SAC | August 2025 - December 2025

- Conducted 10 experimental trials over a 3.05 m track, analyzing translational and rotational work to determine Stirling engine performance.
- Calculated 0.0433 J average mechanical work and 0.0052% experimental efficiency, comparing results with an 86.6% theoretical efficiency and evaluating major energy losses.

Folding Knife Fabrication Project

Mt. SAC | February 2026 - June 2026

- Designed folding components and mechanical interfaces in Fusion 360, applying dimensional tolerances for proper fit, alignment, and assembly.
- Fabricated components from 1084 carbon steel through profiling, beveling, sanding, finishing, and final assembly.
- Performed quenching and tempering of 1084 steel to modify hardness and mechanical properties.

Professional Experiences

Power Tech Engineers Inc. | Intern

December 2022 - March 2023

- Assisted engineers with AutoCAD drafting and review of circuit layouts and technical drawings, applying engineering standards and drafting practices.
- Collaborated with engineers and project teams on technical tasks, project workflows, and documentation.

Lifetastic Learning Inc. | Lead Instructor

June 2023 - May 2026

- Supported over 100 children through individualized educational strategies while collaborating with RBTs, BCBAs, parents, and coworkers.
- Led STEM activities including LEGO, coding, and robotics, while coordinating daily activities and fostering teamwork, problem-solving, and technical skills.

Skills and Awards

SkillsUSA California Regional Competition - Bronze Medalist, Additive Manufacturing | 2022

SkillsUSA California State Competition - Bronze Medalist, Technical Drafting | 2023

SOLIDWORKS CSWP | 2026

CAD: SOLIDWORKS, Fusion 360, AutoCAD, Onshape

Design and Fabrication: 3D Modeling, Mechanical Design, Materials Testing, 3D Printing, Power Tools, Laser Cutting

Programming: MATLAB, Python, C++

Language: Mandarin Chinese, English